SRUSHTI SUDHIR GHATGE

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CAREER OBJECTIVE

T.Y. B.Tech student in Artificial Intelligence & Machine Learning with hands-on experience building explainable ML/DL systems, health-tech applications, and full-stack web projects. Seeking an entry-level opportunity to apply skills in machine learning, deep learning, and software development.

EDUCATION

T.Y. B.Tech — Computer Science and Engineering (Artificial Intelligence & Machine Learning)
D. Y. Patil College of Engineering & Technology, Kasaba Bawada, Kolhapur (Affiliated to Shivaji University)
Expected Graduation: 2027 | Aggregate: 80.16%

Relevant Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Database Management Systems, Information Security, Control Systems

TECHNICAL SKILLS

Languages: Python, SQL, Java, C++, C, PHP, JavaScript

AI / ML / Deep Learning: Scikit-learn, TensorFlow/Keras, OpenCV, SHAP, Grad-CAM, Computer Vision, NLP, Transfer Learning, Feature Engineering

Data Science: Pandas, NumPy, Matplotlib, Seaborn

Web & Databases: Flask, FastAPI, React, HTML5, CSS3, PHP, MySQL, MongoDB

Tools & Cloud: Git/GitHub, VS Code, Google Colab, Jupyter Notebook, Docker, Kubernetes, Microsoft Azure

KEY PROJECTS

NeuroTrack AI — Explainable Multimodal Digital Biomarker Platform *Individual (Ongoing) | Sole Developer*

- Designing a multimodal Clinical Decision Support System tracking digital biomarkers (handwriting + speech) for longitudinal neurological monitoring, explicitly scoped to exclude diagnosis or treatment recommendation
- Building a handwriting-analysis CNN (ResNet/EfficientNet) with Grad-CAM explainability and a speech-analysis pipeline with SHAP-based interpretability
- Designing a deterministic (non-LLM) cross-visit comparison engine and template-based clinical report generator, prioritizing reliability over generative approaches
- Tech: Python, TensorFlow/Keras, OpenCV, Whisper, Librosa, spaCy, SHAP, Grad-CAM, FastAPI, React, PostgreSQL/SQLite

AI RealEstate Architect Platform *Team of 3 | AI/ML Lead*

- Led AI/ML and deep learning architecture components across a 5-module platform unifying floor plan generation, Vastu compliance analysis, energy optimization, cost estimation, and automated reporting
- Designed the floor plan generation approach combining deep learning/generative techniques with practical layout constraints for buildable outputs
- Coordinated technical integration across modules and directed overall technical direction for the team
- Tech: Python, deep learning/generative frameworks, rule-based logic (Vastu/energy heuristics), PDF/report generation

Personalized AI-Based Indian Meal & Insulin Decision-Support System *Individual (Ongoing) | Sole Developer & Researcher*

- Designing a personalized, explainable, safety-first decision-support system using longitudinal CGM, meal, and insulin data in an n-of-1 study design
- Built a novel Personalized Meal Suitability Index (PMSI) ranking Indian meals by safety, nutrition, glucose compatibility, activity, and insulin factors
- Developing a hybrid system combining a rule-based clinical safety engine with an XGBoost-based glucose prediction model, keeping safety-critical decisions rule-governed
- Tech: Python, Scikit-learn/XGBoost, SHAP, rule-based logic engine, FastAPI, React, SQLite

ADDITIONAL PROJECTS

- **Spam & Scam Detection (Multimodal) —** Led a team of 3 to unify text, voice, and image inputs into one classifier using TF-IDF and OCR/Speech-to-Text
- **Mental Health Prediction —** Led preprocessing, feature engineering, model training, and Streamlit deployment for a team of 3
- **Diabetes Prediction —** Built and evaluated a Logistic Regression classifier for diabetes risk prediction
- **Image Captioning —** Built a CNN-LSTM encoder-decoder model to generate captions for input images
- **Furniture E-Commerce Website —** Frontend lead with MongoDB integration for flexible schema design and CRUD-based catalog management

INTERNSHIP

Web Development Intern — VaultofCodes.in *1 Month (Nov 2025)*

- Built login and registration pages as part of a guided web development track
- Developed a cake-baking timer application that displays the live cooking process alongside the countdown

CERTIFICATIONS

- The Ultimate Job Ready Data Science Course — CodeWithHarry (Jul 2026)
- Complete 2026 Python Bootcamp: Learn Python from Scratch — CodeWithHarry (Jul 2026)
- Python Fundamentals for Beginners — Great Learning Academy (Oct 2024)
- Java Programming — Great Learning Academy (Oct 2024)

ACHIEVEMENTS

- Co-authored a copyright-registered literary/dramatic work, "Virtual Vibe: The Ultimate Dress-Up World" — Indian Copyright Office (filed Nov 2024)
- Participated in Poster Presentation Competition 2K24 on Artificial Intelligence & Machine Learning — Dept. of CSE-AIML, D. Y. Patil College of Engineering & Technology

LANGUAGES KNOWN

English, Hindi, Marathi